

Measuring AI Ability to Complete Long Software Tasks: Time Horizon Doubling Every 7 Months

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50% time horizon doubles every ~207 days (~7 months)

A New Metric: 50% Task-Completion Time Horizon

Step 1

Assemble suite

170 software + research tasks spanning 1 second to 8+ hours of skilled-human effort.

Step 2

Measure outcomes

Collect human timings and AI agent success rates across task durations.

Step 3

Fit logistic model

Estimate duration where model reaches 50% success probability.

Methodology figure (paper Figure 2)

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Suite Source	Task Count	Typical Span	Example Tasks
HCAST	97	~1 min–30 hrs	wikipedia_research (1 min), oxdna_simple (9 min), munge_data (56 min)
RE-Bench	7	8 hrs	cuda_backtesting (8 hrs)
SWAA	66	~1–30 sec	find_shell_script (3 sec)

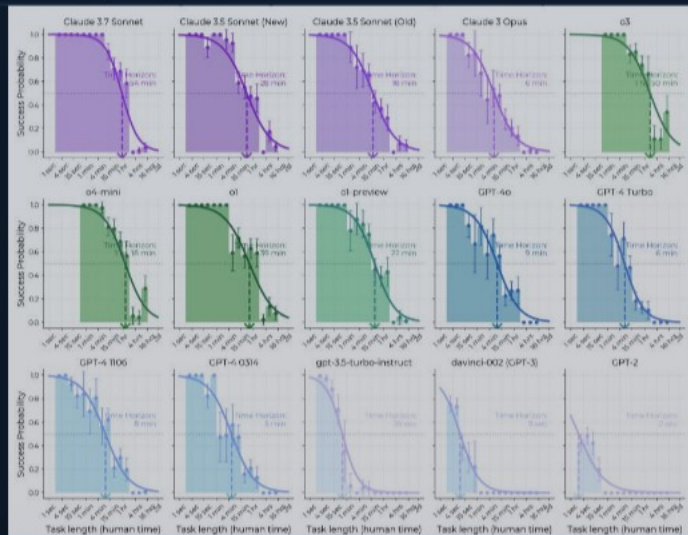
Benchmark depth: combines atomic UI/scripting actions, software engineering workflows, and difficult ML research engineering tasks.

- Coverage ranges from micro-actions (seconds) to sustained engineering tasks (hours).
- Suite is designed to estimate capability-vs-duration curves, not just point scores.
- Directly supports horizon estimation through human-time calibration.

Examples and counts sourced from METR paper brief.

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3) in 6 Years

50% time horizon vs. release date (paper Figure 1)



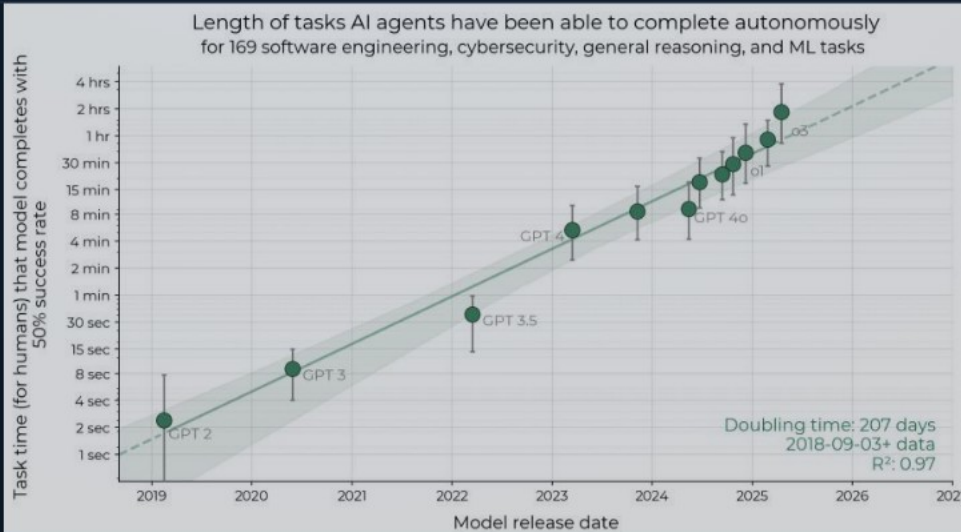
GPT-2: ~2 sec (2019) GPT-4 (0314): ~5 min (2023) o3: ~1h 50m (2025)

Doubling time: ~207 days (~7 months)
Fit quality: $R^2 = 0.97$

- Time horizon increases exponentially across model generations.
- 2023–2025 rate is ~20% faster than the 2019–2025 aggregate trend.
- 80% horizon follows similar shape but is roughly 5× shorter.

Success Probability Drops Sharply as Task Length Increases for All Models

Per-model success probability curves (paper Figure 3)



- All models exhibit declining success as human-task duration increases.
- Frontier curves shift right over time, indicating longer solvable tasks.
- Reference horizons: Claude 3.7 Sonnet (~54 min) vs. o3 (~1h50m).
- Older models remain concentrated in seconds-scale horizons (GPT-2 ~2 sec).

Interpretation: progress is a curve shift, not elimination of failure on long tasks.

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

- **Improved logical reasoning** supports longer sequential plans.
- **Better tool use** increases productivity within coding workflows.
- **Reliability and self-awareness** reduce unforced execution errors.
- **Adaptation to mistakes** enables iterative recovery on complex tasks.

- Limitation:** performance is markedly lower on less structured, "messier" tasks.
- External validity remains open: benchmark may not match real-world labor mix.
 - Supplementary experiments show limited evidence of slower trends on more realistic tasks, but uncertainty persists.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

1-month horizon (167 work hours)

- Naive continuation of observed trend reaches month-long automation between mid-2028 and mid-2031.
- Forecast assumes broadly stable scaling dynamics and benchmark relevance.
- 2023–2025 growth appears ~20% faster than full-period baseline.

Forecast caveats: extrapolation is sensitive to trend breaks, evaluation drift, and real-world distribution mismatch.

- Policy planning should treat this as scenario analysis, not deterministic prediction.
- Risk management timelines may need to compress if acceleration persists.

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

- **Metric strength:** 50% horizon is intuitive and robust ($R^2 = 0.97$ over six years).
- **Growth rate:** capability doubling at ~7 months, with possible post-2024 acceleration.
- **Current frontier:** autonomous completion near 2 hours of skilled-human work (o3).
- **Open validity question:** benchmark tasks may not perfectly represent real-world intellectual labor distributions.
- **Persistent limits:** weaker performance on unstructured tasks and uncertain real-world generalization.